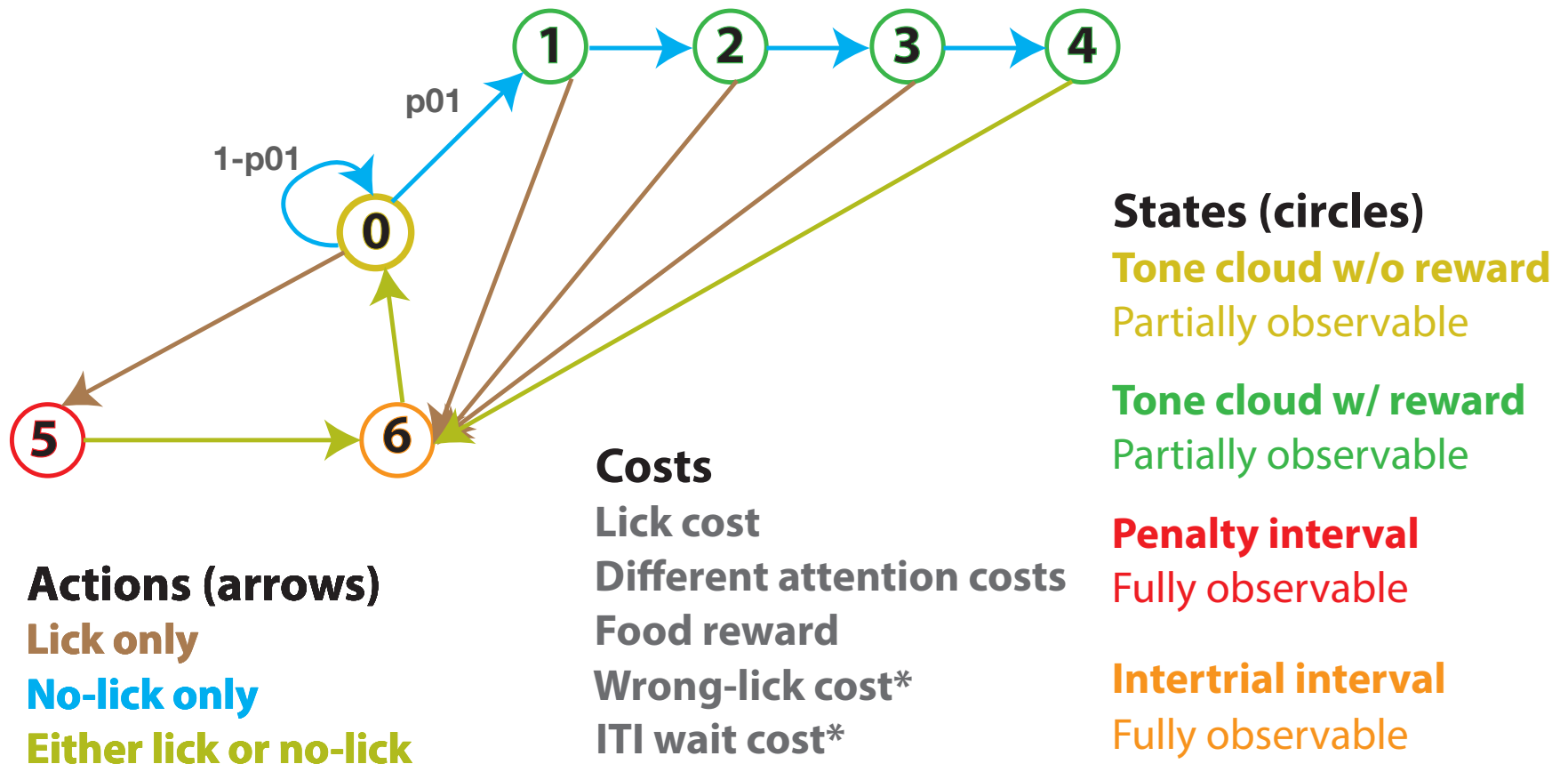


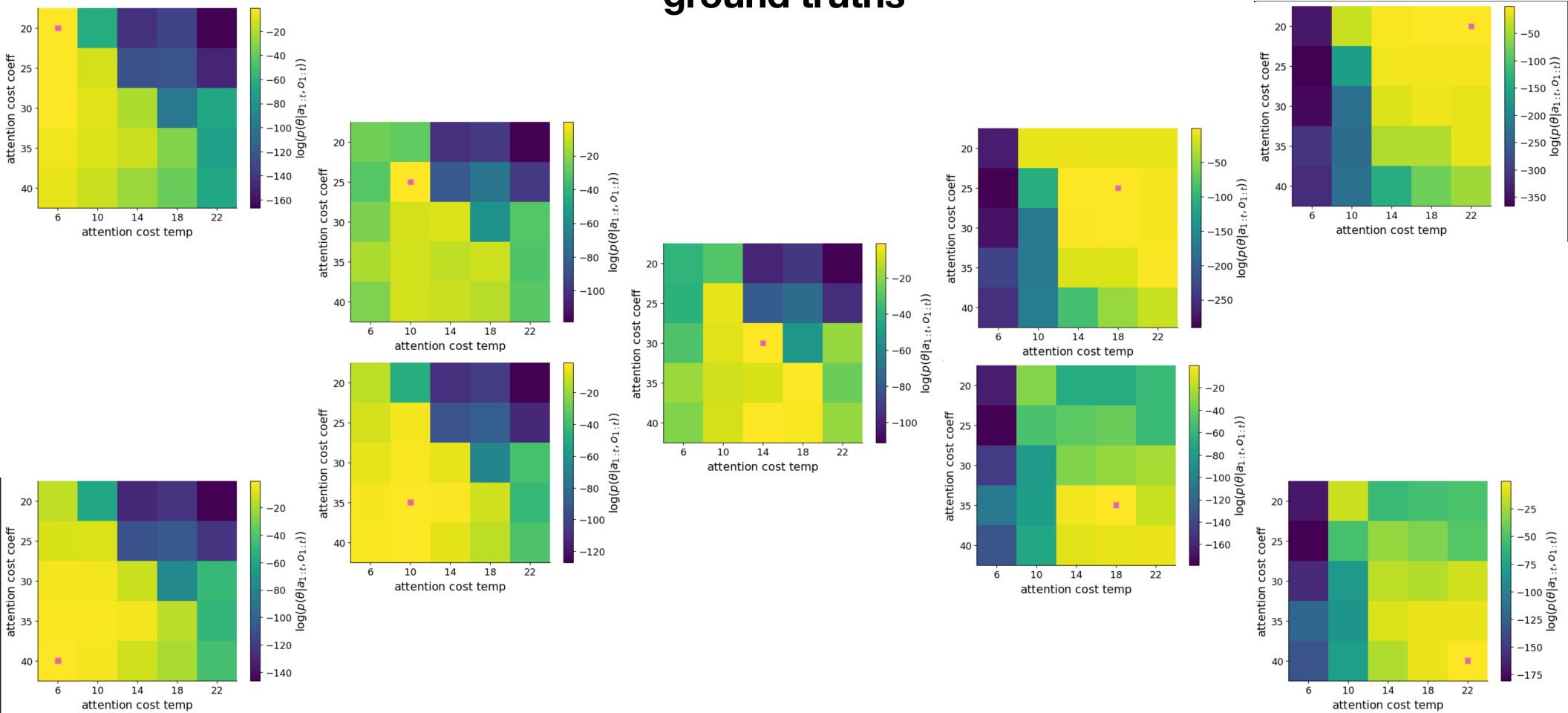
Previously: Auditory foraging (non-episodic)



Auditory foraging (non-episodic)

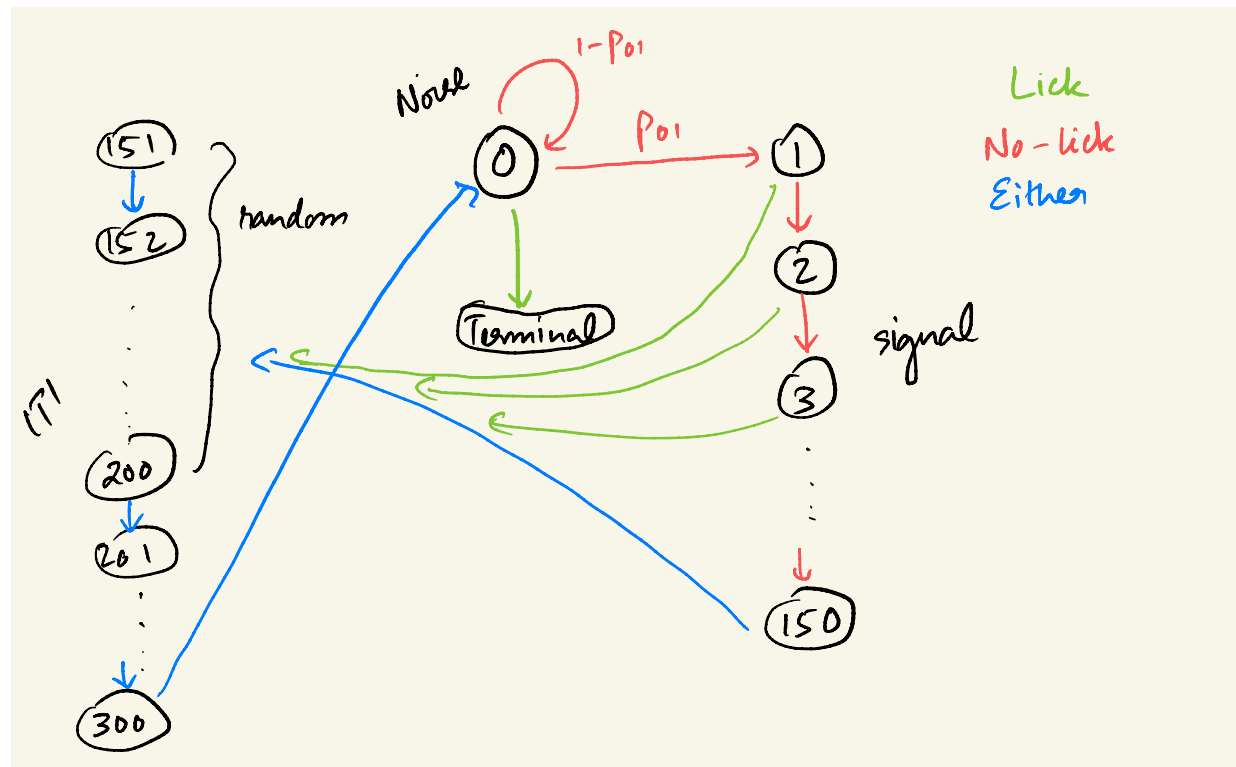
- Inputs - observations, lick choice, attention choice
- Output - cost of attention, parameterized using (coefficient, temperature)

Likelihoods for different ground truths

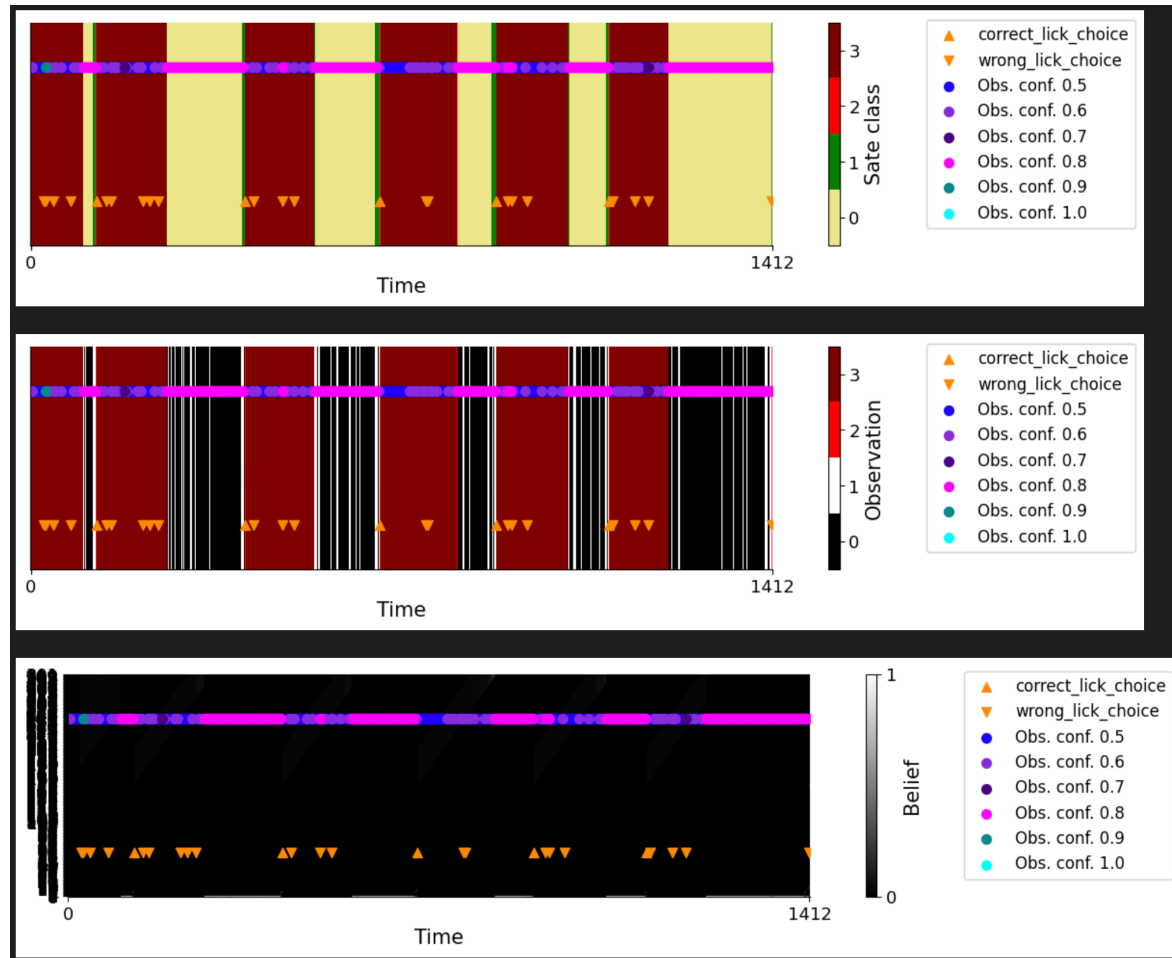


Auditory foraging (episodic)

- Key issue - Good to have 20 milliseconds time resolution to better capture the behavior.
- Solution - episodic case
 - A transition step is 20ms.
 - Episode ends in the first instance of penalty.
 - ITI - uniform [2,3] (150 nodes).
 - Signal duration 3s (150 nodes).
 - Tone cloud, exponential with mean 5s (2 nodes).
 - Episode starts after last instance of penalty.

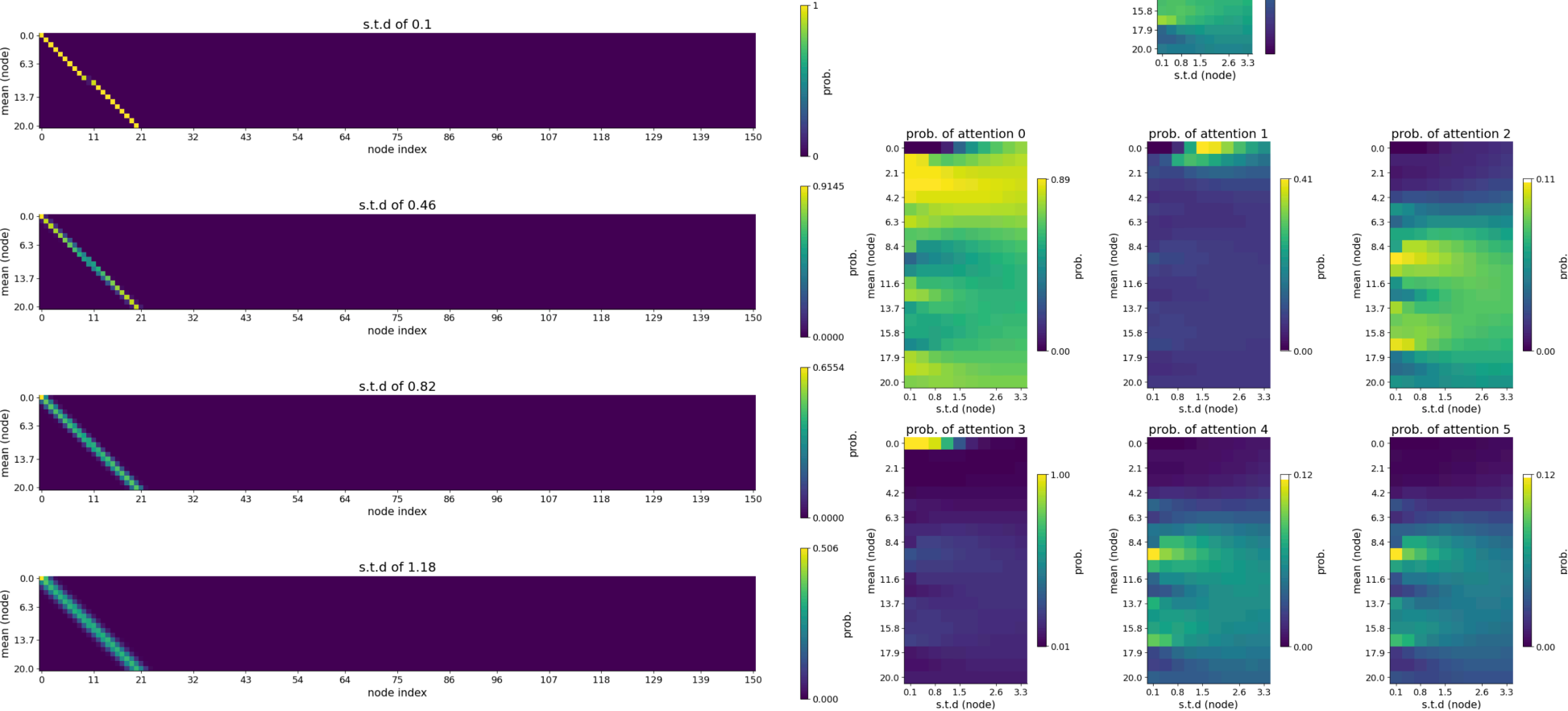


Able to collect some rewards before terminating



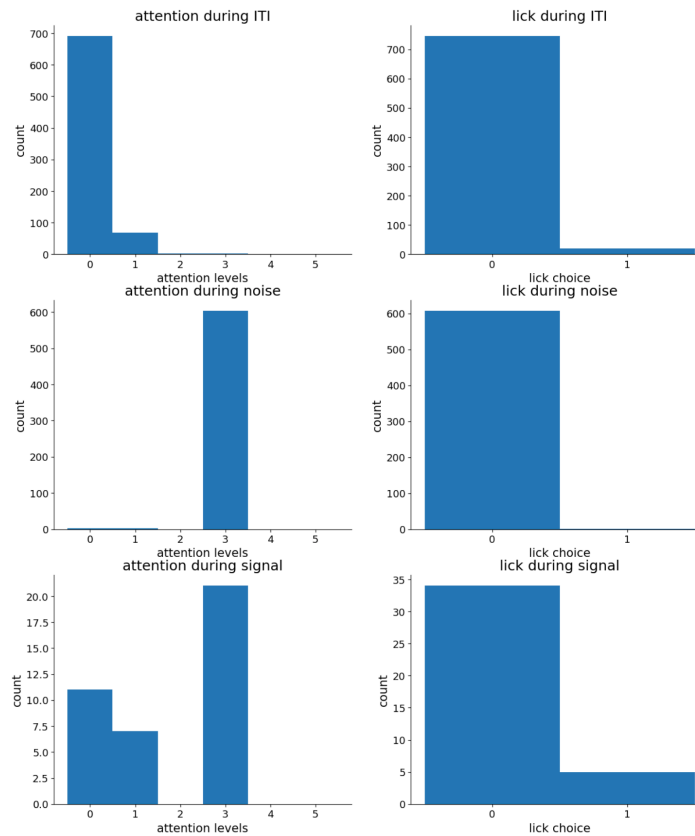
Preliminary analyses before IRC

Artificial beliefs to probe



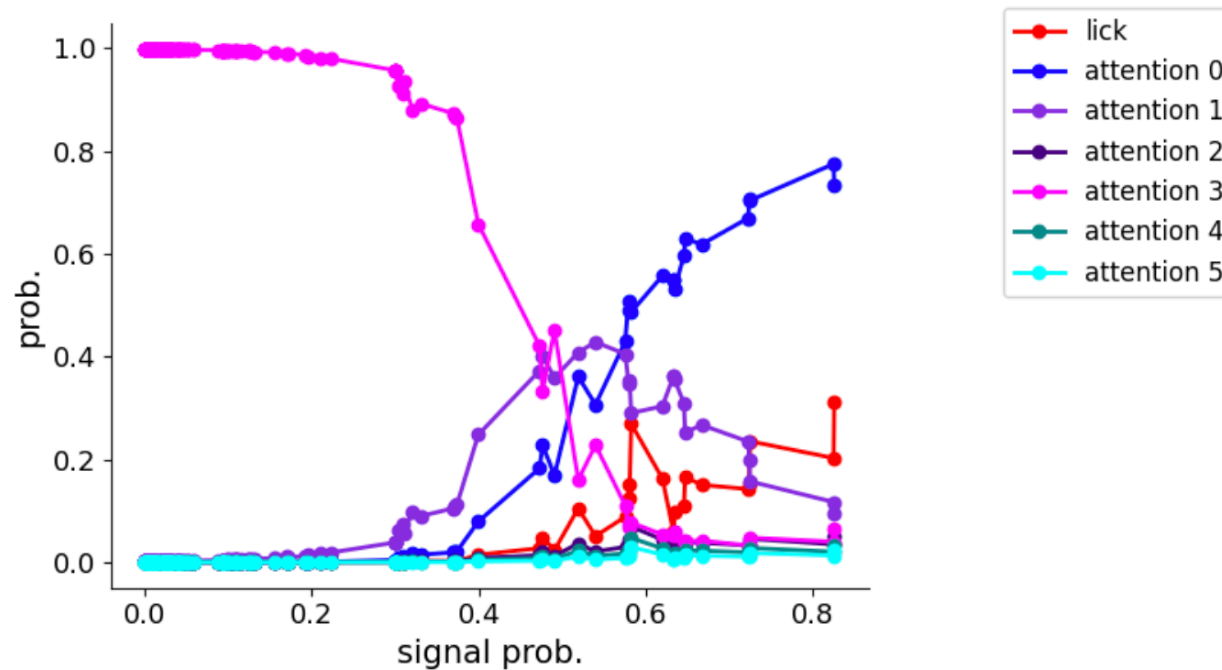
Preliminary analyses

Average action dist. during parts of the episode



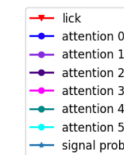
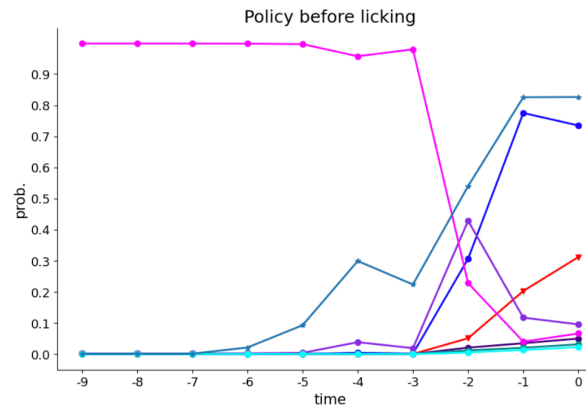
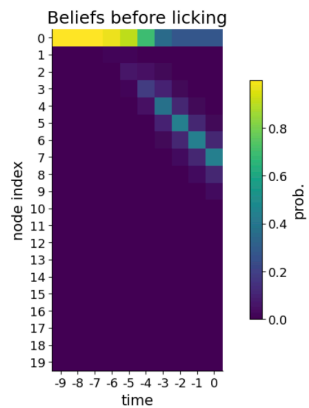
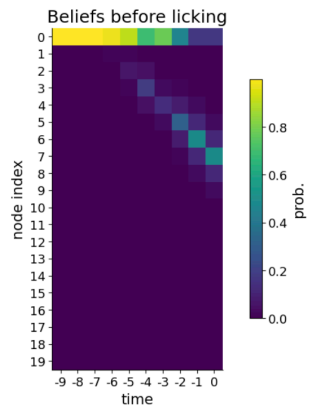
Preliminary analyses

Action dist. as a function of signal probability

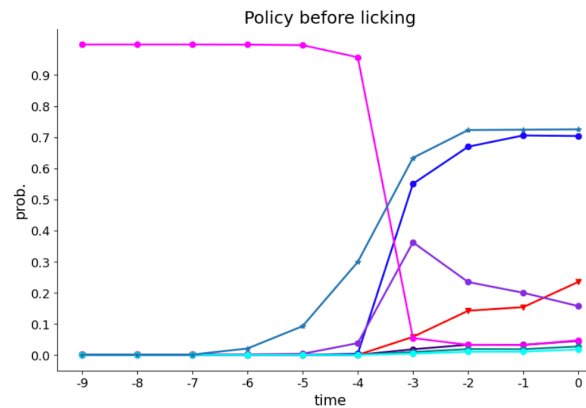


Preliminary analyses

Action dist. before termination (in ordering of time).



Zig-zag effect can be attributed to belief updates.



Takeaway - hard to characterize behavior explicitly.

Currently - Partially observable action

- Inputs - observations, lick choice
- Output - attention choice, cost of attention
- Generating random actions, feeding to IRC, finding which has maximum likelihood
 - e.g. 6 levels of attention, 700 time steps in non-ITI region
 - 6^{700} possibilities worse case. Maybe possible to get rid of very few.
- Different settings of params gives about the same cumulative rewards (5-7). Maybe hard to distinguish just based on lick choices.